

Rosen — §4.1: Divisibility and Modular Arithmetic

Selected Exercises

Alston

Semester 2, 2026

Exercise 5

Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.

Exercise 7

Show that if a , b , and c are integers, where $a \neq 0$ and $c \neq 0$, such that $ac \mid bc$, then $a \mid b$.

Exercise 8

Prove or disprove that if $a \mid bc$, where a , b , and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.

Exercise 17

Show that if n and k are positive integers, then $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.

Exercise 19

Find a formula for the integer with smallest absolute value that is congruent to an integer a modulo m , where m is a positive integer.